

Empirical Realization of Compositional Structure: A Synthesis of the PRINCIPIA PHYSICA Topic Series

PRINCIPIA PHYSICA

Abstract

Three manuscripts of this series treat compositional modelling, Hamiltonian reconstruction, and the empirical limits of structural physics. This paper places them in one frame. We isolate a single mechanism — a linear observation operator with nonzero kernel — and prove that the finite-sample non-identifiability of Hamiltonians and the general finite-data underdetermination theorem are two instances of it. We prove that a monoidal natural transformation between compositional models is invisible to every closed experiment, so that the empirical content of such a model is a gauge class rather than a functor. We identify a third gap, distinct from finite data and from gauge: a local-to-global obstruction measured by first cohomology, of which the non-existence of a global Hamiltonian on the torus and the Aharonov–Bohm phase are structurally parallel instances. We prove that a gauge constraint can destroy tensor factorization outright, answering a standing objection to compositional modelling with a proof rather than a qualification, and record the Groenewold–van Hove theorem as a no-go on functorial transfer between theories. Finally we prove a bound for propagating tolerance through composition, with a counterexample fixing its scope to deterministic composites. The mathematical core is machine-checked in Lean 4 and executed in Haskell. Nothing proved here entails that any model is physically realized, and the paper is organized so that the point at which that entailment fails is explicit.

1 What the series established, and what it did not

The thesis of the treatise is that physics studies empirically realized compositional mathematical structures, and the three manuscripts of this series develop the two halves of that phrase separately on purpose. The compositional paper defines a *model* as a strong symmetric monoidal functor from an interface theory to a system theory (C1) and proves that a model preserves every equation generated by sequential and parallel composition (C2). The Hamiltonian paper proves the regular Legendre correspondence (C3) and that finite sampled trajectories do not pin down an unrestricted Hamiltonian off the sampled set (C4). The empirical-limits paper proves a general underdetermination theorem for classes closed under perturbations vanishing on the tested inputs (C5). The series then records an interpretation (C6): admissibility supplies candidates, realization requires separate evidence.

Read as a sequence these look like three unrelated cautionary results. This paper argues they are not. Sections 2 and 3 show that C4 and C5 are one theorem applied to two observation operators, the reduction being exact rather than analogical. Section 4 shows that a second, logically independent form of non-identifiability — gauge — is present in all three settings, and proves the categorical instance the compositional paper left implicit. Section 5 identifies a third form that none of the papers isolated and that is invisible to the other two. Section 6 records a no-go theorem on transferring compositional structure between theories, and Section 7 supplies the one positive quantitative result the series lacked. Sections 8–10 state the resulting four-status ledger, report what has been machine-checked, and record disagreements and open problems without resolving them.

Epistemic labels are used as in the series: a *definition* stipulates; an *assumption* is granted without proof and carried into everything depending on it; a *theorem*, *proposition*, *lemma*, or *corollary* is proved here; an *established physical result* is attributed and not reproved; an *empirical*

statement has measurement as its warrant; an *interpretation* is a marked reading that no proof here supports; a *conjecture* is open.

2 One frame for the three papers

The three manuscripts share an epistemic pattern, but not a single ambient mathematical type. The reconstruction and finite-record arguments use linear model spaces and observation operators. The compositional argument instead uses categories and functors, which need not carry any natural real-vector-space structure. We therefore state the two vocabularies separately and connect them only when an application explicitly supplies both.

Definition 2.1 (Model class and observation operator). *Let V be a real vector space, $\mathcal{L} \subseteq V$ a linear-model class, D a real vector space of data, and $O : V \rightarrow D$ a linear map, the observation operator. The record of $m \in \mathcal{L}$ is $O(m)$.*

Linearity of O is a restriction, and it is the one that makes the next section work. It holds for the operators used in the linear portions of the series: restriction to a tested set, evaluation of a differential at sampled points, and evaluation of a finite family of linear functionals. It fails for a nonlinear noise model, and Section 10 records that gap. No claim is made that strong monoidal functors form such a subset $\mathcal{L}$ or admit such an O .

Definition 2.2 (Interface theory, system theory, compositional model). *An interface theory is a symmetric monoidal category $(\mathcal{E}, \otimes, I, \sigma)$ whose morphisms are admissible experimental processes; a system theory is a symmetric monoidal category $(\mathcal{M}, \boxtimes, \mathbf{1})$; a compositional model is a strong symmetric monoidal functor $F : \mathcal{E} \rightarrow \mathcal{M}$ with structure isomorphisms $\phi_{X,Y}^F : FX \boxtimes FY \rightarrow F(X \otimes Y)$ and $\phi_0^F : \mathbf{1} \rightarrow F(I)$. This is claim C1 of the series.*

Lemma 2.3 (Substitution). *Let $F : \mathcal{E} \rightarrow \mathcal{M}$ be a functor and $e = e' : X \rightarrow Y$ in $\mathcal{E}$. Then for all composable $a : W \rightarrow X$ and $b : Y \rightarrow Z$,*

$$F(b \circ e \circ a) = F(b) \circ F(e) \circ F(a) = F(b) \circ F(e') \circ F(a).$$

Proof. Equality of morphisms is a congruence for composition, so $b \circ e \circ a = b \circ e' \circ a$ in $\mathcal{E}$. The assignment F makes on each hom-set a function, so equal arguments have equal images. The functor axiom $F(g \circ f) = F(g) \circ F(f)$, applied twice, rewrites each image as the stated composite in $\mathcal{M}$. $\square$

Corollary 2.4 (Contextual invariance). *No context built from sequential composition separates the images under F of processes identified in $\mathcal{E}$.*

Proof. A sequential context is a morphism expression with one hole; filling the hole with e and with e' produces the two sides of Lemma 2.3. $\square$

Lemma 2.3 and Corollary 2.4 are machine-checked; see Section 9.

Definition 2.5 (Closed experiment and normalized scalar). *A state of X is $s : I \rightarrow X$, an effect is $p : X \rightarrow I$, and a closed experiment is a morphism $c \in \mathcal{E}(I, I)$; the experiments $p \circ e \circ s$ are of this form. For a strong monoidal F , the normalized scalar of c is*

$$\langle F, c \rangle = (\phi_0^F)^{-1} \circ F(c) \circ \phi_0^F \in \mathcal{M}(\mathbf{1}, \mathbf{1}).$$

Assumption 2.6 (Probabilistic interpretation). *There is a map $\nu : \mathcal{M}(\mathbf{1}, \mathbf{1}) \rightarrow [0, 1]$ such that the scalars assigned to the mutually exclusive outcomes of each specified measurement sum to one. This is granted, not proved; every probability statement below inherits it.*

Definition 2.7 (Compatibility, identifiability, gauge). *Fix tested inputs T , observed values $\hat{y}_t$, and tolerances $\delta_t \geq 0$. The compatibility region is $C(T, \delta) = \{m \in \mathcal{L} : d(q_t(m), \hat{y}_t) \leq \delta_t \text{ for all } t \in T\}$. The class is identifiable under O when $O(m) = O(n)$ implies $m = n$, and identifiable up to $\approx$ when $O(m) = O(n)$ implies $m \approx n$.*

Proposition 2.8 (Monotonicity of compatibility). *If $T' \subseteq T$ and $\delta_t \leq \delta'_t$ for $t \in T'$, then $C(T, \delta) \subseteq C(T', \delta')$.*

Proof. Let $m \in C(T, \delta)$ and $t \in T'$. Then $t \in T$, so $d(q_t(m), \hat{y}_t) \leq \delta_t \leq \delta'_t$. As t was arbitrary in T' , $m \in C(T', \delta')$. $\square$

Proposition 2.8 unifies the “monotonicity of realization” proposition of the compositional paper with the “tolerance” and “restriction” monotonicity propositions of the empirical-limits paper: the three are one inequality read on two arguments. Both readings are machine-checked (Section 9).

Example 2.9 (Finite stochastic systems). Objects are finite sets, morphisms are column-stochastic matrices, sequential composition is matrix product, parallel composition is the Kronecker product, and ν is the identity on $[0, 1]$. The accompanying Haskell program builds this model and checks that it preserves associativity, the interchange law, and naturality of the symmetry, and that column-stochasticity is closed under both compositions.

3 The kernel theorem: C4 and C5 are one result

Assumption 3.1 (O -closure at m). *There is a subset $K \subseteq \ker O$ with $0 \in K$ and $m + K \subseteq \mathcal{L}$.*

Theorem 3.2 (Kernel underdetermination). *Under Assumption 3.1, every $h \in K$ satisfies $O(m + h) = O(m)$, and the map $h \mapsto m + h$ is injective. Consequently the fibre $\{m' \in \mathcal{L} : O(m') = O(m)\}$ has cardinality at least $|K|$. If $K \neq \{0\}$ then $\mathcal{L}$ is not identifiable under O ; if K is infinite then the record leaves infinitely many admissible models standing.*

Proof. For $h \in K$, linearity of O gives $O(m + h) = O(m) + O(h) = O(m) + 0 = O(m)$, and $m + h \in \mathcal{L}$ by Assumption 3.1, so $m + h$ lies in the fibre. If $m + h = m + h'$ then $h = h'$ by cancellation in V , so the map $h \mapsto m + h$ is injective and the fibre has at least $|K|$ elements. If K contains some $h \neq 0$ then $m + h \neq m$ while $O(m + h) = O(m)$, which contradicts identifiability. $\square$

Theorem 3.2 is short because it is the whole content of both prior non-identifiability results. We now exhibit the reductions.

Corollary 3.3 (C5 is an instance). *Let X be a set, W a real vector space, $V = \{f : X \rightarrow W\}$, $T \subseteq X$ the tested inputs, and $O = R_T$ the restriction $f \mapsto f|_T$. Then R_T is linear, $\ker R_T = \{h : h|_T = 0\}$, and the perturbation-closure assumption of the empirical-limits paper is exactly Assumption 3.1 for a subset $K \subseteq \ker R_T$. The finite-data underdetermination theorem of that paper is therefore Theorem 3.2.*

Proof. Restriction is linear because addition of functions is pointwise: $(f + g)|_T = f|_T + g|_T$ and $(\lambda f)|_T = \lambda(f|_T)$. Its kernel is the set of h with $h(x) = 0$ for all $x \in T$, which is the description of the admitted perturbations in that paper. The conclusion of that paper — that $m + h$ is record-equivalent to m and distinct from it — is the conclusion of Theorem 3.2. $\square$

Corollary 3.4 (C4 is an instance, and the two natural data types agree). *Let (M, ω) be a symplectic manifold of positive dimension, $S = \{z_1, \dots, z_N\} \subset M$ finite, $V = C^\infty(M)$, $D = \bigoplus_{k=1}^N T_{z_k}^* M$, and $O_d(H) = (dH(z_k))_k$. Then:*

1. O_d is linear;
2. the velocity operator $O_X(H) = (X_H(z_k))_k$ satisfies $O_d = \left(\bigoplus_k \omega_{z_k}^\flat\right) \circ O_X$ with each $\omega_{z_k}^\flat$ a linear isomorphism, hence $\ker O_d = \ker O_X$;

3. $\ker O_d$ contains the infinite set $\{\lambda b : \lambda \in \mathbb{R}\}$ for any nonzero smooth bump b supported in an open ball disjoint from S .

Consequently the finite-sample non-identifiability theorem of the Hamiltonian paper is Theorem 3.2 applied to O_d , and the same conclusion holds verbatim for velocity data.

Proof. (1) Exterior differentiation is $\mathbb{R}$ -linear on functions and evaluation at a point is linear, so O_d is a composite of linear maps. (2) With the convention $\iota_{X_H}\omega = dH$, the bundle map $\omega_z^b : T_z M \rightarrow T_z^* M$, $v \mapsto \iota_v \omega_z$, satisfies $\omega_z^b(X_H(z)) = dH(z)$; it is a linear isomorphism at each z because ω is nondegenerate. Applying it fibrewise gives the stated factorization, and precomposition of an injective map preserves kernels, so the two kernels coincide. (3) A finite set in a positive-dimensional manifold has empty interior, so an open coordinate ball U disjoint from S exists; take b nonzero, smooth, supported in U . Each λb vanishes identically on a neighbourhood of every z_k , hence $d(\lambda b)(z_k) = 0$ for all k , and distinct λ give distinct functions because $b \neq 0$. $\square$

Corollary 3.5 (Identification inside a finite linear class). *If $\mathcal{L} = \{\sum_{j=1}^m \theta_j \phi_j\}$ for fixed smooth ϕ_j , then θ is determined by the record if and only if the design map $A\theta = (\sum_j \theta_j d\phi_j(z_k))_k$ has trivial kernel.*

Proof. Two parameters give the same record exactly when their difference lies in $\ker A$, and $\ker A = \{0\}$ is exactly the case $K = \{0\}$ excluded by Theorem 3.2. $\square$

Interpretation 3.6. Corollaries 3.3–3.5 locate non-identifiability in the pair (model class, observation operator) and nowhere else. Enlarging T shrinks $\ker R_T$, but a class closed under bump perturbations retains a nonzero kernel for every finite T , so more data of the same kind is never the remedy. The remedy is a restriction of the class — a degree bound, a dictionary, a smoothness scale — which is a modelling commitment requiring its own warrant. No theorem in this series supplies that warrant.

4 Gauge: what no record can see

Theorem 3.2 concerns a finite record. A second and independent question is what remains invisible to an *ideal, complete* record. We show that all three papers exhibit this second phenomenon, and prove the categorical case.

Theorem 4.1 (Monoidal gauge invisibility). *Let $F, G : \mathcal{E} \rightarrow \mathcal{M}$ be strong monoidal functors and let $\alpha : F \Rightarrow G$ be a monoidal natural transformation, that is, a natural transformation satisfying $\alpha_{X \otimes Y} \circ \phi_{X,Y}^F = \phi_{X,Y}^G \circ (\alpha_X \boxtimes \alpha_Y)$ and $\alpha_I \circ \phi_0^F = \phi_0^G$. Then for every closed experiment $c \in \mathcal{E}(I, I)$,*

$$\langle G, c \rangle = \langle F, c \rangle.$$

No invertibility of α is assumed.

Proof. The unit axiom $\alpha_I \circ \phi_0^F = \phi_0^G$ and invertibility of ϕ_0^F give $\alpha_I = \phi_0^G \circ (\phi_0^F)^{-1}$. Both ϕ_0^F and ϕ_0^G are isomorphisms because F and G are strong, so α_I is an isomorphism with $\alpha_I^{-1} = \phi_0^F \circ (\phi_0^G)^{-1}$. Naturality of α at $c : I \rightarrow I$ is the equation $\alpha_I \circ F(c) = G(c) \circ \alpha_I$, and composing on the right with α_I^{-1} yields $G(c) = \alpha_I \circ F(c) \circ \alpha_I^{-1}$. Substituting into Definition 2.5,

$$\langle G, c \rangle = (\phi_0^G)^{-1} \alpha_I F(c) \alpha_I^{-1} \phi_0^G = (\phi_0^G)^{-1} \phi_0^G (\phi_0^F)^{-1} F(c) \phi_0^F (\phi_0^G)^{-1} \phi_0^G = (\phi_0^F)^{-1} F(c) \phi_0^F = \langle F, c \rangle. \quad \square$$

Corollary 4.2 (The empirical content of a compositional model is a class). *Let $\approx$ be the equivalence relation on strong monoidal functors $\mathcal{E} \rightarrow \mathcal{M}$ generated by “there exists a monoidal natural transformation from one to the other”. Then $F \approx G$ implies $\langle F, c \rangle = \langle G, c \rangle$ for all closed c , and under Assumption 2.6 the two models predict the same probability for every closed experiment.*

Proof. Write $F \sim G$ when $\langle F, c \rangle = \langle G, c \rangle$ for all closed c . The relation $\sim$ is reflexive, symmetric, and transitive because equality of morphisms is. Theorem 4.1 shows that $\sim$ contains every generating pair of $\approx$. An equivalence relation containing the generators of $\approx$ contains $\approx$, because $\approx$ is by construction the smallest such relation. The probability statement is the application of ν to equal scalars, and depends on Assumption 2.6. $\square$

Proposition 4.3 (The Hamiltonian gauge is exactly the constants). *Let (M, ω) be a connected symplectic manifold. If H and K generate the same Hamiltonian vector field, then $H - K$ is constant; conversely adding a constant leaves the field unchanged.*

Proof. If $\iota_X \omega = dH$ and $\iota_X \omega = dK$ for the same X , then $d(H - K) = 0$. A smooth function with vanishing differential on a connected manifold is constant. Conversely $d(H + c) = dH$ for constant c , and ω is nondegenerate, so the generated fields coincide. $\square$

Proposition 4.3 is the continuous statement; its discrete one-dimensional analogue — the kernel of a forward-difference operator is exactly the constants — is machine-checked by induction in the accompanying Lean development, in both directions.

Interpretation 4.4. Sections 3 and 4 isolate two logically independent deficits. The kernel of Theorem 3.2 shrinks as the tested set grows; the gauge of Theorem 4.1 and Proposition 4.3 does not shrink at all, being a degeneracy of the map from mathematics to predictions rather than a shortage of experiments. Conflating them produces two familiar errors: treating a gauge freedom as an experimental target, and treating a genuine data deficit as mere convention.

5 A third gap: local data, global structure

Neither of the previous two mechanisms accounts for the following situation: complete, ideal, local information, and still no determination of the global object. The obstruction is cohomological.

Proposition 5.1 (Symplectic versus Hamiltonian). *Let X be a smooth vector field on a symplectic manifold (M, ω) . Then $\mathcal{L}_X \omega = 0$ if and only if $\iota_X \omega$ is closed, and X admits a global Hamiltonian if and only if $\iota_X \omega$ is exact. The obstruction is the class $[\iota_X \omega] \in H_{\text{dR}}^1(M)$, and it vanishes on every contractible open set.*

Proof. Cartan’s formula gives $\mathcal{L}_X \omega = d(\iota_X \omega) + \iota_X(d\omega)$, and $d\omega = 0$ because ω is symplectic, so $\mathcal{L}_X \omega = d(\iota_X \omega)$; the first equivalence is the vanishing of this expression. A global Hamiltonian is by definition a function H with $\iota_X \omega = dH$, which is exactness. The class $[\iota_X \omega]$ is defined precisely when the form is closed and vanishes precisely when it is exact. On a contractible open set the Poincaré lemma makes every closed one-form exact, so the class vanishes there. $\square$

Counterexample 5.2 (Locally Hamiltonian, globally not). On the two-torus with angular coordinates (q, p) and $\omega = dq \wedge dp$, take $X = \partial_p$. Then $\iota_X \omega = -dq$, which is closed but not exact, since its integral around the q -cycle is $-2\pi \neq 0$ while every exact form integrates to zero over a cycle. Local Hamiltonians exist on every contractible chart; no single-valued global one does.

The physical counterpart of Counterexample 5.2 is not a mathematical curiosity, and it is measured.

Established physical result 5.3 (Aharonov–Bohm). In a field-free but non-simply-connected region surrounding a confined magnetic flux Φ , the relative phase acquired by an electron beam split around the flux is $e\Phi/\hbar$, and the resulting interference shift is observed. The effect was predicted in [12], reported in [13], and confirmed with the field completely shielded by a superconductor in [14]. The gauge-invariant description of the observable is the holonomy of the connection, not the local field strength [15].

Empirical statement 5.4 (What Established result 5.3 contributes here). The local field strength vanishes identically on the accessible region while the observable phase does not. Complete local data therefore fails to determine an observable quantity — an empirical statement, and the reason the corresponding mathematical possibility cannot be dismissed as pathological.

Interpretation 5.5. Counterexample 5.2 and Established result 5.3 are structurally parallel: in each the datum is a closed one-form on a region with $H^1 \neq 0$, and the undetermined quantity is its class. We assert the parallel as a reading, not an identification: the two concern different physical content, are established by different means — one by proof, one by measurement — and neither is derived from the other here. What it does establish is that the local-to-global gap is a third mechanism, independent of Sections 3 and 4: it survives complete local data, and unlike gauge it concerns a quantity that is observed.

A related objection is that gauge constraints can prevent decomposing a physical state space into local tensor factors. It is correct, and the following makes it precise.

Proposition 5.6 (A constraint can destroy factorization). *Let $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ and let $\mathcal{H}_{\text{phys}} = \{\psi \in \mathcal{H} : (Z \otimes Z)\psi = \psi\}$, where $Z = \text{diag}(1, -1)$. Then $\dim \mathcal{H}_{\text{phys}} = 2$, and there are no subspaces $V_1, V_2 \subseteq \mathbb{C}^2$ with $\mathcal{H}_{\text{phys}} = V_1 \otimes V_2$.*

Proof. In the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ the operator $Z \otimes Z$ is diagonal with entries $1, -1, -1, 1$, so its $+1$ eigenspace is $\text{span}\{|00\rangle, |11\rangle\}$, of dimension 2. Suppose $\mathcal{H}_{\text{phys}} = V_1 \otimes V_2$. Then $\dim V_1 \cdot \dim V_2 = 2$ with each $\dim V_i \leq 2$, so one factor has dimension 1 and the other dimension 2. Assume $\dim V_1 = 1$, say $V_1 = \mathbb{C}v$ with $v = \alpha|0\rangle + \beta|1\rangle \neq 0$, and $V_2 = \mathbb{C}^2$; the other case is identical after exchanging the tensor factors. Then $v \otimes w \in \mathcal{H}_{\text{phys}}$ for every $w \in \mathbb{C}^2$. Writing $v \otimes w = |0\rangle \otimes (\alpha w) + |1\rangle \otimes (\beta w)$ and comparing with the general element $a|00\rangle + b|11\rangle$ of $\mathcal{H}_{\text{phys}}$, membership forces $\alpha w \in \mathbb{C}|0\rangle$ and $\beta w \in \mathbb{C}|1\rangle$. Take $w = |0\rangle + |1\rangle$, which is nonzero. Then $\alpha w \in \mathbb{C}|0\rangle$ forces $\alpha = 0$, and $\beta w \in \mathbb{C}|1\rangle$ forces $\beta = 0$, so $v = 0$, contradicting $v \neq 0$. Hence no such V_1, V_2 exist. $\square$

Interpretation 5.7. Constraints of exactly the type in Proposition 5.6 — a Gauss law — are what distinguishes a gauge theory’s physical state space from the tensor product of local kinematic spaces; the resulting difficulty for local decomposition is documented in [16]. The consequence for Definition 2.2 is a scope restriction, not a refutation: a symmetric monoidal structure on $\mathcal{M}$ guarantees that a tensor product *exists in the category* and guarantees nothing about whether a constrained physical state space is such a product of local factors. Inferring the latter from the former is what this treatise forbids, and Proposition 5.6 shows the inference is not merely unjustified but false.

6 A no-go on functorial transfer

Sections 3–5 concern what a model fails to determine. There is a further limit: structure-preserving maps *between* theories need not exist at all.

Established physical result 6.1 (Groenewold–van Hove). There is no linear map Q from the Poisson algebra of polynomials on $\mathbb{R}^{2n}$ to self-adjoint operators satisfying all of $Q(\{f, g\}) = \frac{1}{i\hbar}[Q(f), Q(g)]$ for all polynomials f, g ; $Q(1) = \text{Id}$; and irreducibility of the representation generated by $Q(q^i), Q(p_i)$. The obstruction is present already at low degree. The result is due to [17, 18]; a modern statement and proof, with the precise subalgebras on which quantization does and does not exist, is [19].

Interpretation 6.2. We do not reprove Established result 6.1, and draw one consequence for Definition 2.2. Compositional modelling is often defended on the ground that a structure-preserving map between theories transports results without loss; Established result 6.1 shows that the existence of such a map is a substantive question with a negative answer in a central case. Compositionality is therefore a constraint, not a convenience: it can be so strong that nothing satisfies it. This

qualifies claim C2, which is conditional on a functor existing — exactly what is denied for the classical-to-quantum case in full generality.

Conjecture 6.3 (Characterizing admissible transfers). *There is a characterization, in terms of the interface theory alone, of which sub-theories of a classical Poisson theory admit a symmetric monoidal functor into a quantum system theory preserving the stated brackets and irreducibility. Partial results are known for specific subalgebras [19]; we state the general question as open and claim no progress on it here.*

7 Propagating tolerance through composition

The series proves that composition preserves equations and that finite data underdetermine models, but it supplies no quantitative statement about how a realization claim for parts becomes a realization claim for the whole. The following supplies one, and the counterexample that follows fixes its scope.

Lemma 7.1 (Composition bound). *Let $(A, d_A), (B, d_B), (C, d_C)$ be metric spaces, let $S \subseteq A$, and let $f_1, \widehat{f}_1 : A \rightarrow B$ and $f_2, \widehat{f}_2 : B \rightarrow C$. Assume*

1. $d_B(f_1(a), \widehat{f}_1(a)) \leq \varepsilon_1$ for all $a \in S$;
2. f_2 has a nondecreasing modulus of continuity μ , so that $d_B(b, b') \leq t$ implies $d_C(f_2(b), f_2(b')) \leq \mu(t)$;
3. $d_C(f_2(b), \widehat{f}_2(b)) \leq \varepsilon_2$ for all $b \in \widehat{f}_1(S)$.

Then for every $a \in S$,

$$d_C(f_2(f_1(a)), \widehat{f}_2(\widehat{f}_1(a))) \leq \mu(\varepsilon_1) + \varepsilon_2.$$

Proof. Fix $a \in S$ and insert the intermediate point $f_2(\widehat{f}_1(a))$. The triangle inequality in C gives

$$d_C(f_2(f_1(a)), \widehat{f}_2(\widehat{f}_1(a))) \leq d_C(f_2(f_1(a)), f_2(\widehat{f}_1(a))) + d_C(f_2(\widehat{f}_1(a)), \widehat{f}_2(\widehat{f}_1(a))).$$

Hypothesis (1) gives $d_B(f_1(a), \widehat{f}_1(a)) \leq \varepsilon_1$, so hypothesis (2) bounds the first term by $\mu(\varepsilon_1)$. Hypothesis (3), applied at the point $\widehat{f}_1(a) \in \widehat{f}_1(S)$, bounds the second term by ε_2 . $\square$

Corollary 7.2 (Lipschitz and chain forms). *If $\mu(t) = Lt$ the bound reads $L\varepsilon_1 + \varepsilon_2$. For a chain $f_n \circ \dots \circ f_1$ compared with $\widehat{f}_n \circ \dots \circ \widehat{f}_1$, where f_i is L_i -Lipschitz and stage i has tolerance ε_i , the composite tolerance is at most*

$$E_n \leq \sum_{i=1}^n \left(\prod_{j=i+1}^n L_j \right) \varepsilon_i.$$

Proof. The Lipschitz case is Lemma 7.1 with $\mu(t) = Lt$. For the chain, argue by induction on n . For $n = 1$ the bound is ε_1 , which is hypothesis (1). Assume $E_{n-1} \leq \sum_{i=1}^{n-1} (\prod_{j=i+1}^{n-1} L_j) \varepsilon_i$. Applying Lemma 7.1 with inner maps the two $(n-1)$ -fold composites, outer maps f_n and $\widehat{f}_n$, and $\mu(t) = L_n t$ gives $E_n \leq L_n E_{n-1} + \varepsilon_n$. Substituting the inductive bound and distributing L_n across the sum produces $\sum_{i=1}^{n-1} (\prod_{j=i+1}^n L_j) \varepsilon_i + \varepsilon_n$, which is the stated expression, the $i = n$ term having empty product 1. $\square$

Both bounds are met with equality on the examples in the accompanying Haskell program, so a wrong constant would be detected rather than hidden by slack. Lemma 7.1 is machine-checked in Lean in an integer modulus form.

Proposition 7.3 (Components do not determine the composite). *There exist two joint distributions on $\{0, 1\}^2$ with identical single-component marginals and different probability for the event that the two outputs agree.*

Proof. Let P be uniform on $\{0, 1\}^2$ and let Q give probability $\frac{1}{2}$ to each of $(0, 0)$ and $(1, 1)$. Each marginal of P is uniform, and each marginal of Q is uniform because $Q(\text{first} = 0) = \frac{1}{2}$ and $Q(\text{second} = 0) = \frac{1}{2}$. The event that the outputs agree has probability $\frac{1}{2}$ under P and 1 under Q . $\square$

Assumption 7.4 (Compositional closure). *Predictions for a composite are determined by the component models together with the declared interface, with no unmodelled shared cause, memory channel, or context dependence across that interface.*

Interpretation 7.5. Lemma 7.1 propagates a *deterministic* scope claim on metric hypotheses alone. Proposition 7.3 shows that no analogue holds for joint statistics without a further premise, the componentwise data being identical in the two cases while a composite prediction differs. The premise that closes the gap is Assumption 7.4, and it is empirical: tests for cross-component correlation, memory, leakage, and contextual dependence bear on it, whereas the availability of a tensor product in $\mathcal{M}$ does not.

8 The four-status ledger

The synthesis of the three papers is that a claim in structural physics occupies exactly one of four statuses, and that the series’ apparent variety of negative results is the variety of ways of confusing them.

Status	Established by	Fails to give
Mathematical admissibility	Proof inside a formal system	Any empirical claim
Finite-record compatibility	Data plus a stated tolerance	Uniqueness (Thm. 3.2)
Identifiability in a class	Trivial kernel (Cor. 3.5)	Warrant for the class
Physical realization	Measurement, within a stated scope	Extrapolation beyond scope

Table 1: The four statuses. Each row takes a different kind of warrant, and no upward inference is valid without an added premise stated here as an assumption.

Interpretation 8.1. Table 1 is the content of claim C6, refined. The negative results distribute across the rows: Theorem 3.2 blocks the step from row two to row three; Theorem 4.1 and Proposition 4.3 show that row three is at best a statement about gauge classes; Section 5 shows that even ideal local data leaves row four underdetermined when $H^1 \neq 0$; and Established result 6.1 shows that row one can itself be empty for a demanded structure. So do the positive ones: Lemma 2.3 lives in row one, Corollary 3.5 gives a criterion for row three, and Lemma 7.1 propagates only a mathematical compatibility bound for deterministic composites. It does not propagate row four. Physical realization of a composite additionally requires Assumption 7.4, including the empirical claim that connection introduces no unmodelled back-reaction, loading, memory, or shared cause.

9 Formal artifacts and what they do and do not certify

Two machine artifacts accompany this paper.

`formal/lean/Proofs.lean` is a Lean 4 development [20] using the core library only, with no Mathlib dependency. It formalizes identifiability and identifiability up to a gauge relation (Definition 2.7); record congruence and the kernel theorem in finite-record form (Theorem 3.2), with an explicit witness — two integer-valued models on $\mathbb{N}$ agreeing on the tested list $[0, 1, 2, 3]$ and differing at 7; the statement that a difference operator’s kernel is exactly the constants, proved by induction in both directions (the discrete analogue of Proposition 4.3); monotonicity of compatibility in the tolerance and in the tested set (Proposition 2.8); the composition bound in an integer modulus form (Lemma 7.1); and the substitution lemma with its contextual-invariance corollary (Lemma 2.3,

Corollary 2.4). Lean 4.33.0 checks the file with no errors or warnings. An axiom audit reports dependence on `propext` and `Quot.sound` only, with the two categorical results depending on no axioms; no result invokes `Classical.choice`.

`formal/haskell/Core.hs` is a Haskell program depending on `base` only. It builds the stochastic-matrix model of Example 2.9 and checks functoriality; constructs the bump perturbation of Corollaries 3.3 and 3.4 and confirms that it leaves the record and its first derivatives on the tested set unchanged while separating the models by $e^{-1} \approx 0.3679$ off it; confirms that a degree bound restores identification; checks the additive gauge of Proposition 4.3; integrates the harmonic oscillator with a Störmer–Verlet scheme, reporting a maximum relative energy drift of 2.5×10^{-5} over 4×10^4 steps; performs the identification of Corollary 3.5 and confirms that the degenerate design at $q = 0$ is refused; checks Proposition 2.8; and verifies Lemma 7.1, Corollary 7.2, and Proposition 7.3, the first two with the bound attained exactly. It type-checks under `-Wall` with no warnings; all 21 checks pass.

Interpretation 9.1. Machine checking certifies that the proofs of the stated *mathematical* propositions are correct. It certifies nothing about realization, and cannot: the Lean development contains no predicate that could express realization, and every numerical input to the Haskell program is a stipulated literal marked in the source as synthetic. The final Lean theorem makes this explicit — any realization predicate constant on record-equivalence classes must assign the same status to two models the record cannot separate, a statement about the impotence of the formalism rather than a discovery about nature.

10 Recorded disagreements, failures, and open problems

Per the epistemic contract of the series, we record rather than resolve.

An unresolved review verdict. An independent review of the topic series returned the verdict REJECT at confidence 0.95, with three blocking findings: that the compositional manuscript identified tensor admissibility with physical realization while ignoring gauge constraints; that the Hamiltonian manuscript presented a reconstruction existence claim without proof or citation; and that the empirical-limits manuscript stated parameter scaling bounds as empirical facts without empirical input. That review was directed at the drafts, and the revised manuscripts address the second finding with a proved existence criterion. We take no position on whether the revisions discharge the verdict, because the review has not been rerun against them. What this paper adds is independent of that question: Proposition 5.6 proves the substance of the first finding rather than qualifying it, and this synthesis asserts no empirical scaling bound of the kind at issue in the third.

Counterexamples supplied by review and retained. The review offered the Aharonov–Bohm effect and the Groenewold–van Hove theorem as counterexamples to the series. Both are retained here as substantive limits rather than deflected: the first as Established result 5.3 and Empirical statement 5.4, the second as Established result 6.1. Neither is reproved here.

Failed constructions. Three attempts did not succeed. First, we found no proof that componentwise realization implies composite realization, and Proposition 7.3 shows none exists without Assumption 7.4. Second, Lemma 7.1 has no counterpart for statistical models under total variation or any other declared distance absent an independence or closure premise; we record its absence rather than a weakened version. Third, we did not obtain a version of Theorem 3.2 for nonlinear observation operators: the proof uses linearity essentially, at $O(m + h) = O(m) + O(h)$.

Open problems. The three conjectures of the series remain open and are not advanced here: modular identifiability of monoidal composites; structured recovery of a Hamiltonian in a fixed

dictionary under a restricted-injectivity condition; and hierarchical propagation of validated uncertainty through composition. To these we add Conjecture 6.3, and the question of whether the cohomological gap of Section 5 admits a formulation in the language of Definition 2.1, which would unify all three mechanisms rather than the two that Section 3 unifies; we have no such formulation.

11 Conclusion

The three manuscripts prove different theorems about the same gap. Theorem 3.2 shows that two of their central negative results are one result about a linear map with nonzero kernel; Theorem 4.1 supplies the categorical form of a second, independent degeneracy; Section 5 exhibits a third that neither of the first two detects; and Proposition 5.6 turns the standing objection from gauge constraints into a theorem. Lemma 7.1 is the sole positive propagation result, and Proposition 7.3 bounds its scope. Claim C6 emerges unchanged and better supported.

Bibliographic note. Categorical background [1, 2, 3]; geometric mechanics [4, 5, 6, 7]; under-determination [8, 9, 10, 11]; physical inputs [12, 13, 14, 15, 16, 17, 18, 19]; proof assistant [20]. Established physical results are attributed here, not reproved.

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